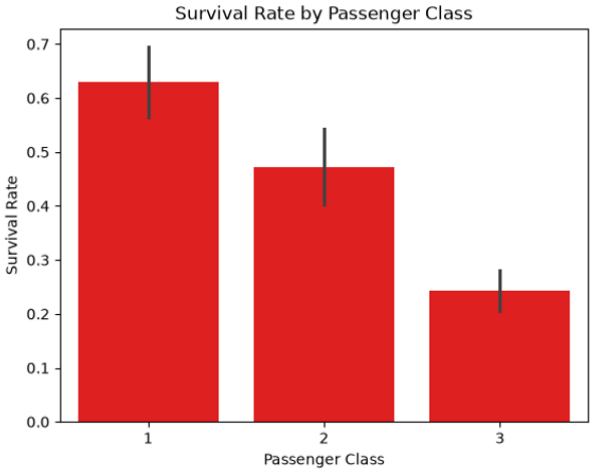
Lesley Lin 100458187

Assignment 2 – Titanic Data Analysis Final Report

Hypothesis One:
Survival Rate vs. Passenger Class

The survival rate was strongly associated with passenger class. First-class passengers had the highest survival rate at 63% (mean = 0.63), followed by second-class passengers at 47% (mean = 0.47), while third-class passengers had the lowest survival rate at 24% (mean = 0.24).

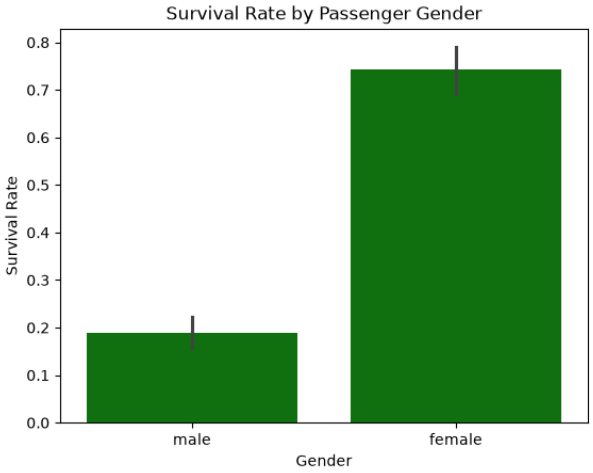
The bar chart shows that first-class passengers had significantly better survival rates.



Hypothesis Two:
Survival Rate vs. Passenger Gender

The survival rate was strongly associated with gender. Female passengers had a significantly higher survival rate of 74% (mean = 0.74), whereas male passengers had a survival rate of only 19% (mean = 0.19).

The bar chart visually highlights a stark contrast between the two groups.



Hypothesis Three:
Survival Rate vs. Age

The survival rate was not strongly associated with age overall. The two groups share very similar mean ages (30.6 years for non-survivors vs. 28.3 years for survivors).

While the dual histograms show largely similar distributions overall, minor differences appear in specific ranges, such as the younger age groups (0–20). Therefore, age did not serve as a dominant overall factor in survival.

